Orthopedic Surgery and Biomechanics

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Orthopedic surgery

- Acute/Planned
- Patients with chronic diseases/ ”healthy” patients
- Extremities and spine
- Different tissues
  - → Bone
  - → Muscle
  - → Tendons
  - → Ligaments
  - → Nerves
  - → Vessels
- Infants/adults
- 600 g – 100 kg
- 30 cm – 2 m

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Orthopedic surgery

- Acute
  - → Trauma
    - Fractures
    - Wounds
    - Nerve injury
    - Disturbed circulation
  - → Infections
    - Osteomyelitis
    - Septic artrhritis
- Semi acute
  - → Tumours

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Orthopedic surgery

- Planned
  - → Deformity correction
    - Congenital
    - Aquired
  - → Osteoarthritis
  - → Activity limiting problems
    - Sports medicin/traumatology
- Consulting
- Surgery
- Follow-up

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Pediatric Orthopedics

- Orthopaedia=straight child
  - → Orthos=straight
  - → Paideia=child
- Pediatric orthopedics
  - → The growing individual
  - → The growing skeleton
  - → Normal growth and development

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Aims for today

- Orthopedic surgery changing biomechanics
  - → Different tissues
  - → Different technics
  - → No full coverage!
  - → A smorgasbord!
- Orthopedic surgery (and biomechanics) is fun!
- Importent meeting between engineers and surgeons
- Give a stimulous to new ideas

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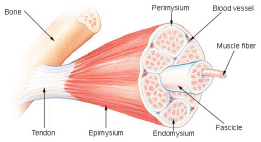
Outline for today

- Soft tissue
  - → Muscles
  - → Tendons
- Joints
  - → Replacement
  - → Fusion
- Bones
  - → Osteotomies
    - Lower extremity
    - Upper extremity
- Questions are allways welcome!

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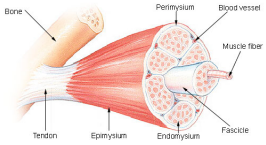
Soft tissue surgery

- Muscles and tendons
  - →Muscles
    - Origin
    - Insertion
    - Force generator
  - →Tendons
    - Connects muscle to bone
    - Converts power to motion



Muscles

- Myotomy
  - →Adductors
    - Scissoring in CP
- Myofasciotomy
  - →Gastrocnemius
    - Spastic equinus in CP
    - Idiopathic toe walking



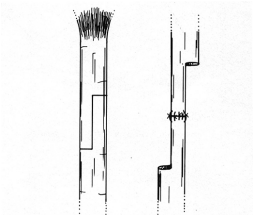
Tendons

- Tenotomy
  - →Psoas tendon
    - CP
  - →Achilles tendon
    - Clubfoot



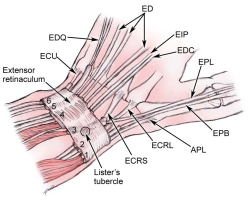
Tendons

- Tendon lengthening
  - →Achilles tendon in CP
  - →Achilles tendon in idiopathic toe-walking
    - Z-lengthening (open)
    - Fractional lengthening (percutaneous)



Tendon

- Tendon transfer
  - →Replace ruptured tendon
    - EIP->EPL
  - →Nerve injury



Tendons

- Tendon transfer
  - →Balancing an unbalaced joint
    - Tibialis anterior



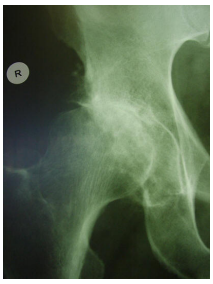
Joint replacement

- Normal hip



Joint replacement

- Hip osteoarthritis
  - →Reduced height of cartilage
  - →Joint space narrowing
  - →Subchondral sclerosis
  - →Osteophytes



Joint replacement

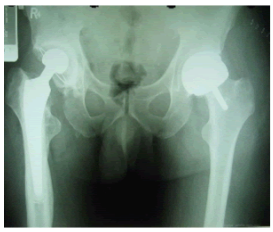
- Hip arthroplasty
- First in 19th century
- Popular in the 1970´s
- Sweden 2024:
  - 23.000 surgeries
  - +5000 due to fracture



Sir John Charnley 1911-1982

Joint replacemnet

- Total hip replacement
- Hip resurfacing



Joint replacement

- Normal knee



Joint replacement

- Knee osteoarthritis
  - →Reduced height of cartilage
  - →Joint space narrowing
  - →Translation
  - →Sweden 2024: 21.000 surgeries



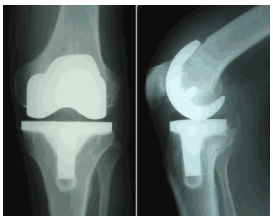
Joint replacement

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Knee artroplasty



X-ray



Joint replacement

Karolinska Institutet

Shoulder



Reverse shoulder



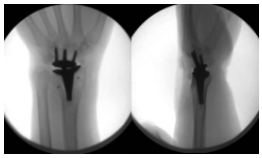
Joint replacement

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Elbow



Wrist



Joint replacement

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Finger



Ankel



Joint replacement

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???

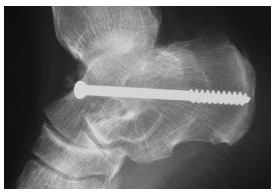


Arthrodesis

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- Ankylosis
- Joint ossification
- Pain
  - → Osteoarthritis
  - → Rheumatoid artritis
- Dislocation
  - → Neuromuscular disease
- Salvage procedure
  - → Arthroplasty failure

Subtalar arthrodesis

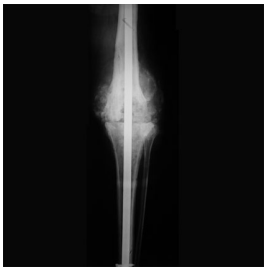


Arthrodesis

Hip



Knee



Correction osteotomy

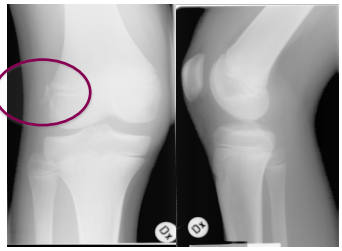
- Varus/Valgus
- Extension/Flexion
- Rotation
- Lengthening/Shortening
- Combinations

Correction osteotomy

- Open wedge
- Closed wedge

Distal femoral osteotomy

- 14 years old boy
- Knee wrench
- Epifphyseal bridge



Distal femoral osteotomy

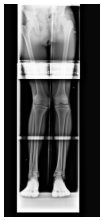
- Increasing valgus



Distal femoral osteotomy

- Increasing valgus





Distal femoral osteotomy

- Open wedge
- Plate and screws



Distal femoral osteotomy

1 month



12 months



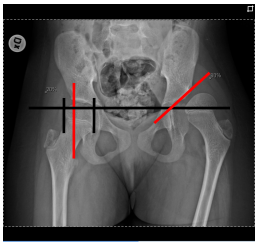
Proximal femoral osteotomy

- Hip migration
- Neuromuscular disease
  →Cerebral palsy
  →Myelomeningocele
  →Rett´s syndrome



Proximal femoral osteotomy

- Reimer´s index
  →Migration %
- Acetabular index
  → acetabular angle



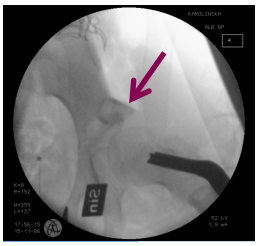
Proximal femoral osteotomy

- Varus osteotomy
- Closed wedge
- Reduction of migration
- Plate and screws
- Extension if needed
- Rotation if needed



Pelvcic osteotomy

- Often combined with femoral osteotomy
- Decrease of acetabular index
- Increase in acetabular coverage
- Recycling of bone graft!



Femoral and pelvic osteotomy

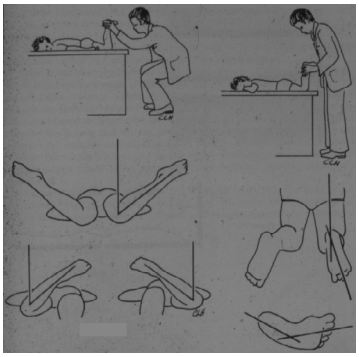
6 weeks post op



Rotational osteotomy

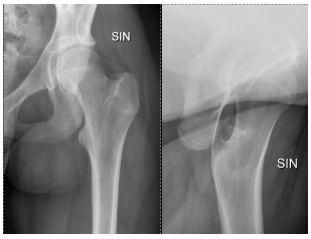
- Intoeing
- Level:
  →Hip?
  →Femur?
  →Tibia?
  →Foot?

Rotational osteotomy



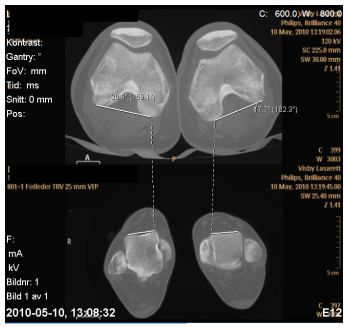
Rotational osteotomy

- Conventional x-ray



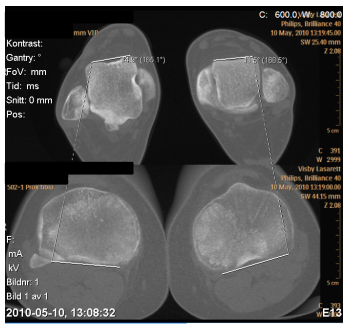
Rotational osteotomy

- CT
- Knee-Ankle



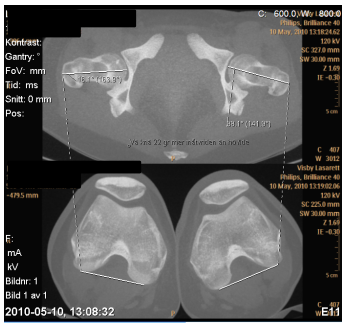
Rotational osteotomy

- CT
- Prox/Dist Tibia



Rotational osteotomy

- CT
- Hip-Knee



Rotational osteotomy

- Diaphyseal osteotomy
- Correction of rotation
- Fixation with intramedullary nail and locking nails



Leg length discrepancy

- Is the long leg longer or the short leg shorter?
- Increasing or stable discrepancy?

- Acquired discrepancy
  - → Fracture
  - → Infection
  - → Circulation
- Congenital discrepancy
  - → Hemihypertrophy
  - → Dysmelia
  - → Idiopathic

Leg length discrepancy

- Treatment options
  →Extend the short leg
  - Leg lengthening
    →Allowes simultaneous correction of deformity

  →Shorten the long leg

  →Inhibit growth in the long leg
  - Epiphyseodesis

Leg lengthening

- Posteromedial bowing of the tibia
- Change of ROM in the ankle



Leg lengthening

- Posteromedial bowing of the tibia
- Change of ROM in the ankle
  →Increased dorsiflexion
  →Decreased plantarflexion



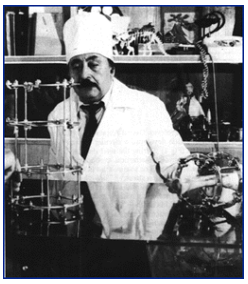
Leg lengthening

- Remodulation
  →Decreasing angulation
- Increasing leg lenght discrepancy
  →6 cm



Leg lengthening

- Distraction osteogenesis
- Callus distraction
- Gavril Ilizarov
  - 1921-1992
  - Kurgan, Siberia



Leg lengthening

- Correction in all directions
- Mid diaphyseal osteotomy



Leg lengthening

- TSF
- Taylor Spatial Frame
- Web based software



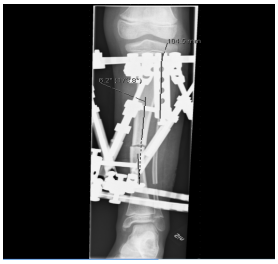
Prescription

Date	Day	Strut 1	Strut 2	Strut 3	Strut 4	Strut 5	Strut 6
11/18/12	0	91	92	112	92	92	93
11/19/12	1	92	93	113	93	93	94
11/20/12	2	93	94	114	94	94	95
11/21/12	3	94	95	115	95	95	96
11/22/12	4	95	96	115	96	96	96
11/23/12	5	95	97	116	97	97	97
11/24/12	6	96	98	117	98	97	98
11/25/12	7	97	99	118	98	98	99
11/26/12	8	98	100	119	99	99	100
11/27/12	9	99	101	120	100	100	101
11/28/12	10	100	102	120	101	101	102
11/29/12	11	101	103	121	102	102	102

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Leg lengthening

7 days



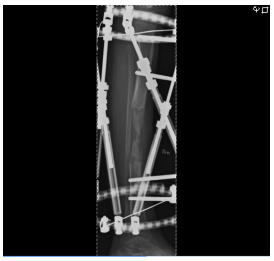
50 days



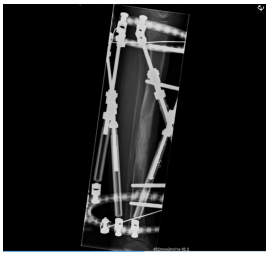
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Leg lengthening

4 months



8 months



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Leg lengthening

- Equal leg lenght
- Joint surfaces of the proximal and distal tibia are parallel



Leg lengthening

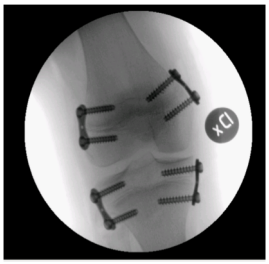
- Intramedullar lengthening
- FitBone ®
- Telescopic nail
- Reciever
- Control unit



Epiphyseodesis

- Inhibit growth by closing the growth plate
  →Permanent
  - Physeal ablation

  →Temporary
  - 8-plate
  - Staples



8-plate

- Temporary epiphyseodesis
- Minor surgery
- No restrictions in physical activities



8-plate

- Fracture of the distal tibia and fibula
- Open reducion and internal fixation
  →Plates and screws





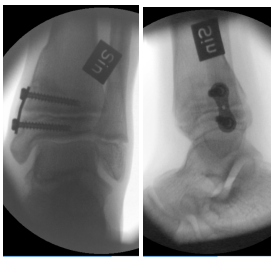
8-plate

- Partial closure of the growth plate
  →Bone bridge
- Increasing valgus deformity



8-plate

Post op

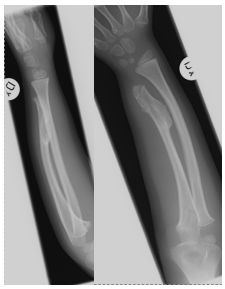


6 months



Upper extremity

- Heriditary osteocartilaginous exostosis



Heriditary osteocartilaginous exostosis

- Increasing deformity
- Growth inhibition
- And what´s even worse…



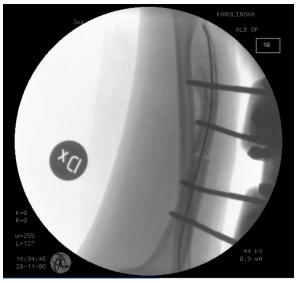
Heriditary osteocartilaginous exostosis

- Luxation of the radial head
  - →Decreased ROM
    - Flexion/Extension
    - Pronation/Supination

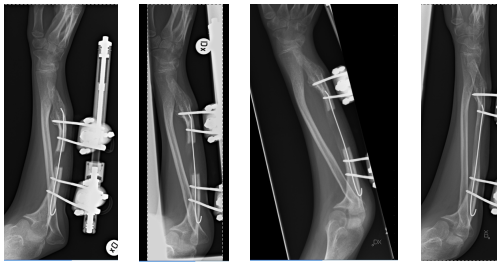


Heriditary osteocartilaginous exostosis

- Callus distraction
- Monotube



Heriditary osteocartilaginous exostosis



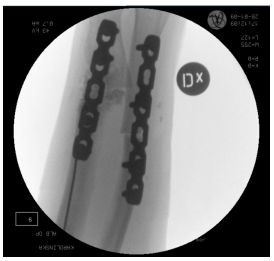
Heriditary osteocartilaginous exostosis

- Longer, but not straight
- Reduction of the radial head
- Increased ROM



Heriditary osteocartilaginous exostosis

Post op



4 months



Heriditary osteocartilaginous exostosis

- A sucecss story?
- Complications are not rare…
- You need patience to deal with these patients!



Summary

- Orthopedic surgery (and biomechanics) is fun!
- Thera are many established ways of handeling orthopedic and biomechanical problems
- There is always a need for invention!
- Surgeons and engineers need each other in this area
- "There are no difficult surgeries, just surgeries that take long time and surgeries you have never done before"

Jacob Lykke Olesen

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Thank you!